Ownership Transfers and Eviction Filing Rates

Philly Evictions Project

2026-03-28

Table of contents

1 Overview **1**

2 Main Filing Results **2**

3 Filing Heterogeneity by Acquirer Type **4**

4 Occupancy, Demographics, Complaints, and Permits **5**

5 Rent Outcomes **9**

6 Rental Evidence **11**

7 Appendix: Output Inventory **13**

1 Overview

This note reports the streamlined annual transfer-evictions specification. The main outcome is `filing_rate = num_filings / total_units` in a full building-year panel from 2006 to 2019, with single-transfer treated buildings compared against buildings never transferred in the RTT data. All annual event studies are estimated on `t \in [r event_window_min, r event_window_max]`, with omitted event time `t = r event_ref_period`.

For filing outcomes, the main specification is a weighted stacked event study:

$$y_{iat} = \sum_{k \in \{-3,-1,0,1,2,3\}} \beta_k \mathbf{1}\{\text{rel_time}_{ia} = k\} + \alpha_{ia} + \lambda_{at} + \varepsilon_{iat},$$

estimated on a cohort-stacked panel with weights `q_weight × total_units`, where `q_weight` is the Wing et al. corrective control weight and `\alpha_{ia}` and `\lambda_{at}` are stack-specific parcel and year fixed effects. Non-filing outcomes continue to use the annual `PID + year` event-study design on the same `-3:3` window.

Operationally, `q_weight` corrects the fact that the same never-transferred controls are reused in multiple cohort stacks. Multiplying by `total_units` then makes the corrected stacked estimator unit-weighted, so the filing results are a weighted average over unit exposure rather than a simple parcel average.

The annual surface has been pruned to four blocks:

- filing-rate event studies
- occupancy, resident-composition, complaint, and permit outcomes
- rent outcomes using the unified and safe series
- rental-evidence event studies for `1-unit` and `2+ unit` buildings

Portfolio filing types are computed using the helper logic from the production script, with portfolio size and filing rates defined using all holdings, including `1-unit` buildings. The filing and neighborhood-outcome regressions then restrict the estimation sample as needed.

The acquirer bins are:

- `Single-purchase`: leave-one-out portfolio exposure is exactly `0` unit-years.
- `Small portfolio`: leave-one-out portfolio exposure is positive but below 100 unit-years.
- `Low-filer portfolio`: leave-one-out portfolio exposure is at least 100 unit-years and the leave-one-out filing rate is below 0.05 filings per unit-year.
- `High-filer portfolio`: leave-one-out portfolio exposure is at least 100 unit-years and the leave-one-out filing rate is at least 0.05 filings per unit-year.

For the filing-rate outcome and for the leave-one-out filing-rate bins, annual filing rates are capped at 0.75 filings per unit-year before aggregation.

2 Main Filing Results

The main filing figures now use the weighted stacked estimator directly. The filing sample grid separates `1-unit` properties from the multifamily stock. The main justification for the `2+ unit` analysis sample is that the `2+` and `5+` estimates are close, while `1-unit` properties are shown separately as the single-family contrast and the `10+` cut is treated as a stricter appendix validation rather than the main specification.

At `t = +1`, the filing-rate coefficient is 0.0088 for `2+ units`, 0.0051 for `5+ units`, and 0.0044 for `10+ units`.

Table 1: Sample sizes for the streamlined annual filing-rate event studies

Sample	Treated transfers	Control PIDs	Stacked rows
All buildings	32,006	123,376	6,960,484
1-unit	24,676	90,413	5,148,966
2+ units	7,330	32,976	1,811,495
2+ units, rental evidence	6,148	28,792	985,907
5+ units	662	5,089	258,330
5+ units, rental evidence	548	3,874	113,778
10+ units	290	2,551	121,631
10+ units, rental evidence	236	1,923	57,728

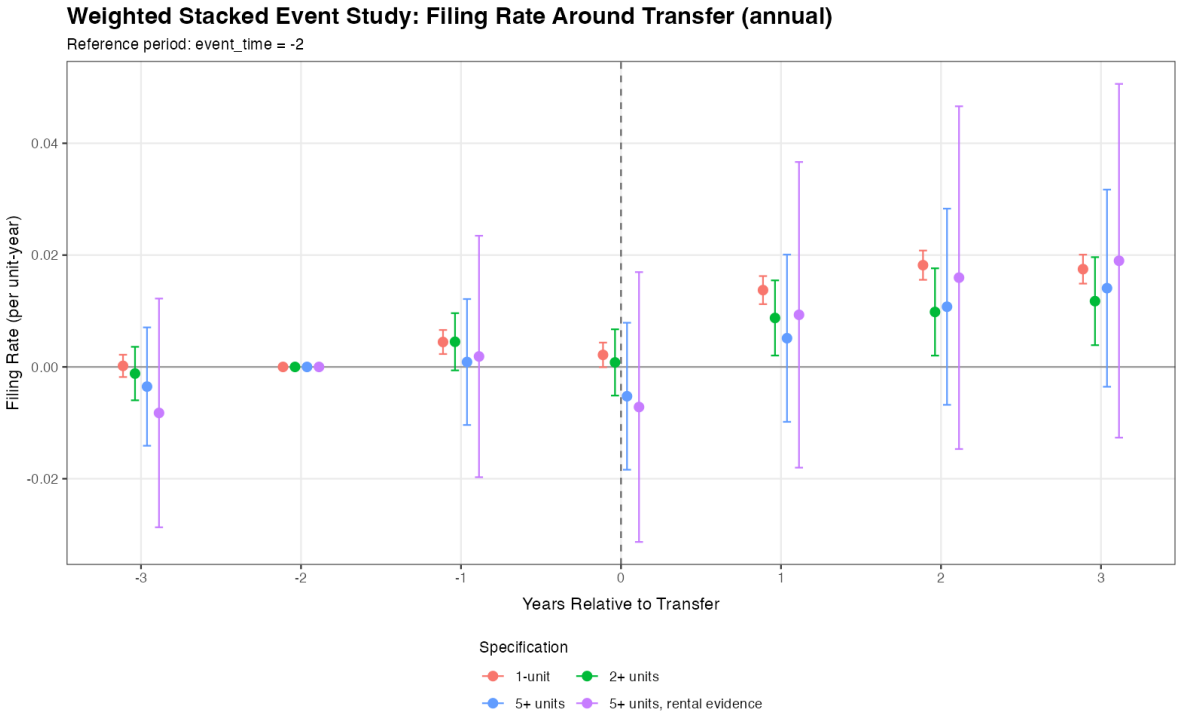

Figure 1: Weighted stacked annual filing-rate event study by sample restriction. The main figure overlays 1-unit, 2+, 5+, and 5+ rental-evidence samples; 10+ remains a stricter appendix check in the tables.

					filing_rate			
Sample	All buildings	1-unit	2+ units	2+ units, rental evidence	5+ units	5+ units, rental evidence	10+ units	10+ units, rental evidence
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
st_et_m3	-0.0006	0.0002	-0.0012	-0.0057	-0.0035	-0.0082	-0.0043	-0.0107
	(0.0013)	(0.0010)	(0.0024)	(0.0054)	(0.0054)	(0.0104)	(0.0067)	(0.0125)
st_et_m1	0.0045***	0.0045***	0.0045*	0.0103*	0.0009	0.0019	0.0013	0.0013
	(0.0014)	(0.0011)	(0.0026)	(0.0060)	(0.0057)	(0.0110)	(0.0071)	(0.0132)
st_et_0	0.0015	0.0022*	0.0008	-0.0019	-0.0052	-0.0072	-0.0094	-0.0152
	(0.0016)	(0.0011)	(0.0030)	(0.0065)	(0.0067)	(0.0123)	(0.0083)	(0.0148)
st_et_1	0.0112***	0.0137***	0.0088**	0.0083	0.0051	0.0093	0.0044	0.0079
	(0.0019)	(0.0013)	(0.0034)	(0.0077)	(0.0076)	(0.0140)	(0.0094)	(0.0166)
st_et_2	0.0138***	0.0182***	0.0098**	0.0068	0.0108	0.0160	0.0096	0.0145
	(0.0021)	(0.0013)	(0.0040)	(0.0087)	(0.0090)	(0.0156)	(0.0110)	(0.0187)
st_et_3	0.0144***	0.0175***	0.0118***	0.0084	0.0141	0.0190	0.0138	0.0190
	(0.0022)	(0.0013)	(0.0040)	(0.0088)	(0.0090)	(0.0161)	(0.0111)	(0.0194)
Observations	6,960,484	5,148,966	1,811,495	985,907	258,330	113,778	121,631	57,728
R^2	0.44893	0.26811	0.60936	0.63480	0.79893	0.79218	0.81745	0.80832
Within R^2	9.24×10^{-5}	0.00011	7.81×10^{-5}	5.8×10^{-5}	0.00021	0.00028	0.00026	0.00039
pid_cohort fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
yr_cohort fixed effects	✓	✓	✓	✓	✓	✓	✓	✓

Table 2: Selected filing-rate coefficients from the streamlined sample grid

Sample	t+1	t+2	t+3
1-unit	0.0137	0.0182	0.0175
10+ units	0.0044	0.0096	0.0138
10+ units, rental evidence	0.0079	0.0145	0.0190
2+ units	0.0088	0.0098	0.0118
2+ units, rental evidence	0.0083	0.0068	0.0084
5+ units	0.0051	0.0108	0.0141
5+ units, rental evidence	0.0093	0.0160	0.0190

3 Filing Heterogeneity by Acquirer Type

The main heterogeneity split uses four acquirer types:

- `High-filer portfolio`
- `Low-filer portfolio`
- `Small portfolio`
- `Single-purchase`

These categories are defined from the helper-function leave-one-out portfolio measures, while the regression itself is restricted to `2+ unit` buildings. In the current annual weighted stacked outputs, the `t = +2` coefficient is 0.0325 for high-filer portfolios and -0.0069 for low-filer portfolios.

The standard errors differ sharply across these bins. At `t = +2`, the high-filer estimate has standard error 0.0436 versus 0.0068 for low-filer acquirers, which largely reflects the much smaller treated sample for high-filer acquisitions.

Table 3: Weighted stacked 2+ filing-rate sample sizes by acquirer type

Acquirer type	Treated transfers	Control PIDs	Stacked rows	Trimmed cohorts
High-filer portfolio	320	32,976	1,764,405	8
Low-filer portfolio	886	32,976	1,767,856	8
Small portfolio	1,572	32,976	1,772,560	8
Single-purchase	4,484	32,976	1,792,808	8

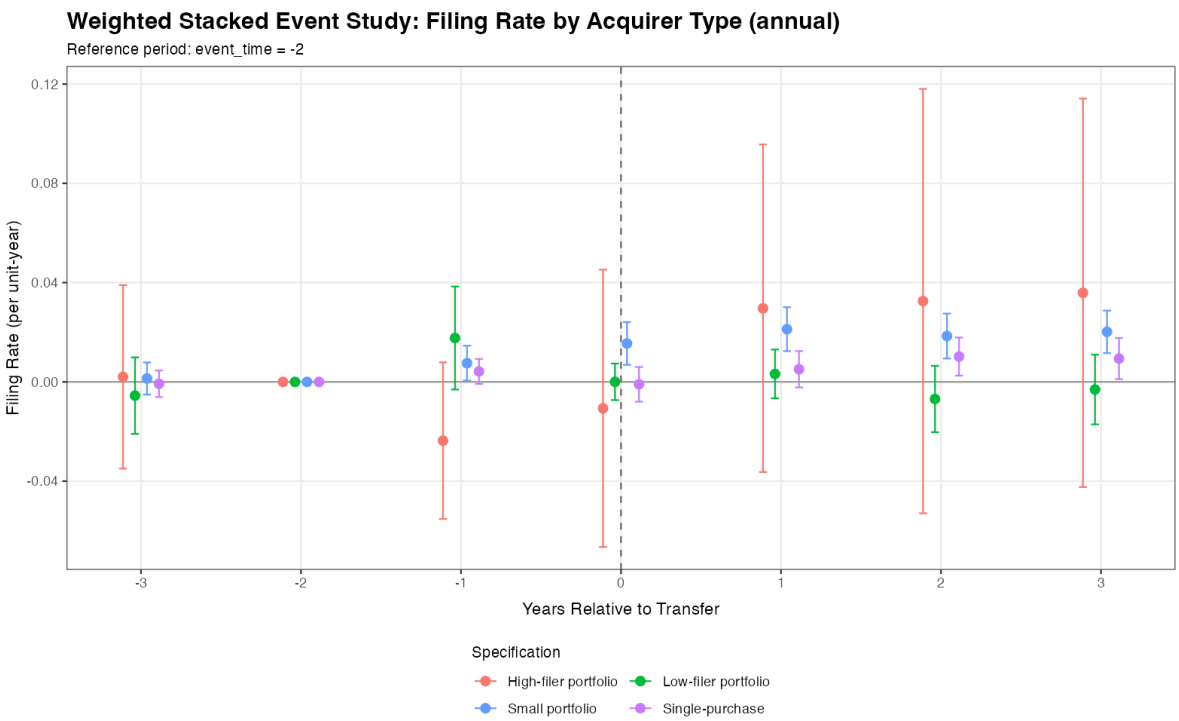

Figure 2: Annual filing-rate event study for 2+ unit buildings, split by acquirer filing type. Controls are never-transferred buildings in the same 2+ unit sample.

	filing_rate			
Acquirer type	High-filer portfolio	Low-filer portfolio	Small portfolio	Single-purchase
	(1)	(2)	(3)	(4)
st_et_m3	0.0020	-0.0055	0.0014	-0.0008
	(0.0189)	(0.0079)	(0.0033)	(0.0027)
st_et_m1	-0.0237	0.0177*	0.0076**	0.0042
	(0.0161)	(0.0106)	(0.0036)	(0.0026)
st_et_0	-0.0106	2.62×10^{-5}	0.0155***	-0.0010
	(0.0285)	(0.0038)	(0.0044)	(0.0036)
st_et_1	0.0296	0.0032	0.0212***	0.0051
	(0.0337)	(0.0050)	(0.0045)	(0.0037)
st_et_2	0.0325	-0.0069	0.0185***	0.0102***
	(0.0436)	(0.0068)	(0.0046)	(0.0039)
st_et_3	0.0359	-0.0031	0.0202***	0.0094**
	(0.0399)	(0.0072)	(0.0044)	(0.0042)
Observations	1,764,405	1,767,856	1,772,560	1,792,808
R^2	0.61259	0.61097	0.61044	0.61078
Within R^2	0.00010	2.99×10^{-5}	3.7×10^{-5}	4.01×10^{-5}
pid_cohort fixed effects	✓	✓	✓	✓
yr_cohort fixed effects	✓	✓	✓	✓

Table 4: Selected 2+ unit filing-rate coefficients by acquirer type

Acquirer type	t+1	t+2	t+3
High-filer portfolio	0.0296	0.0325	0.0359
Low-filer portfolio	0.0032	-0.0069	-0.0031
Single-purchase	0.0051	0.0102	0.0094
Small portfolio	0.0212	0.0185	0.0202

4 Occupancy, Demographics, Complaints, and Permits

These outcomes are estimated on the `2+ unit` sample and split by the same acquirer-type bins as the filing regressions. The output surface here is intentionally narrower than the earlier report: the focus is occupancy, resident composition, complaints, and permits.

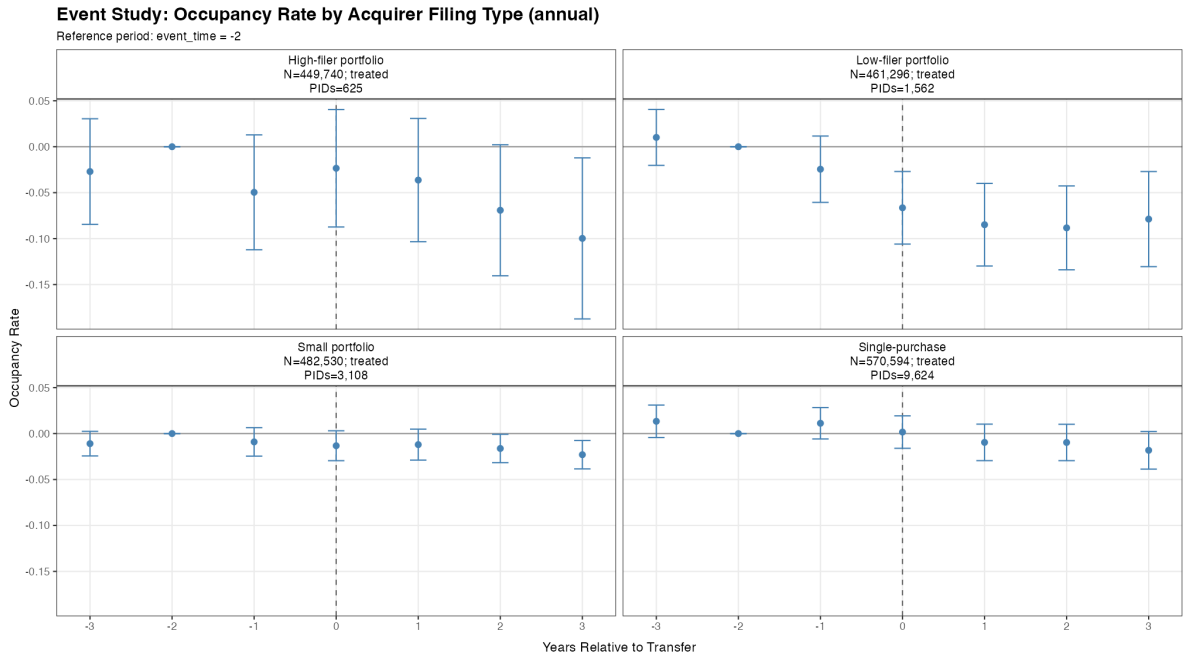

Figure 3: Occupancy-rate event study for 2+ unit buildings, overall and by acquirer filing type.

	occupancy_rate			
Acquirer type	High-filer portfolio	Low-filer portfolio	Small portfolio	Single-purchase
	(1)	(2)	(3)	(4)
et_m3	-0.0270	0.0101	-0.0110	0.0133
	(0.0293)	(0.0155)	(0.0068)	(0.0090)
et_m1	-0.0496	-0.0245	-0.0091	0.0112
	(0.0319)	(0.0184)	(0.0079)	(0.0087)
et_0	-0.0234	-0.0664***	-0.0133	0.0016
	(0.0326)	(0.0202)	(0.0083)	(0.0090)
et_1	-0.0363	-0.0849***	-0.0120	-0.0096
	(0.0342)	(0.0229)	(0.0086)	(0.0101)
et_2	-0.0692*	-0.0883***	-0.0163**	-0.0097
	(0.0364)	(0.0233)	(0.0079)	(0.0101)
et_3	-0.0998**	-0.0788***	-0.0230***	-0.0182*
	(0.0447)	(0.0264)	(0.0079)	(0.0104)
Observations	449,740	461,296	482,530	570,594
R^2	0.49400	0.50014	0.49623	0.50568
Within R^2	0.00037	0.00096	4.74×10^{-5}	0.00016
PID fixed effects	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓

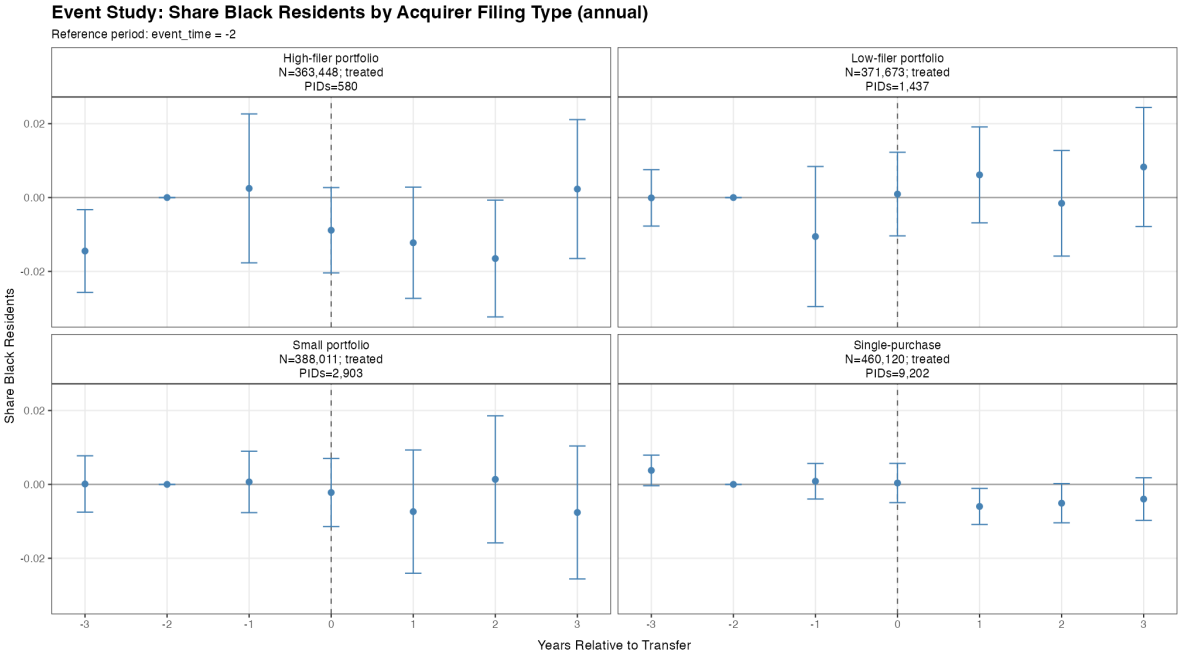

Figure 4: Share Black residents for 2+ unit buildings, overall and by acquirer filing type.

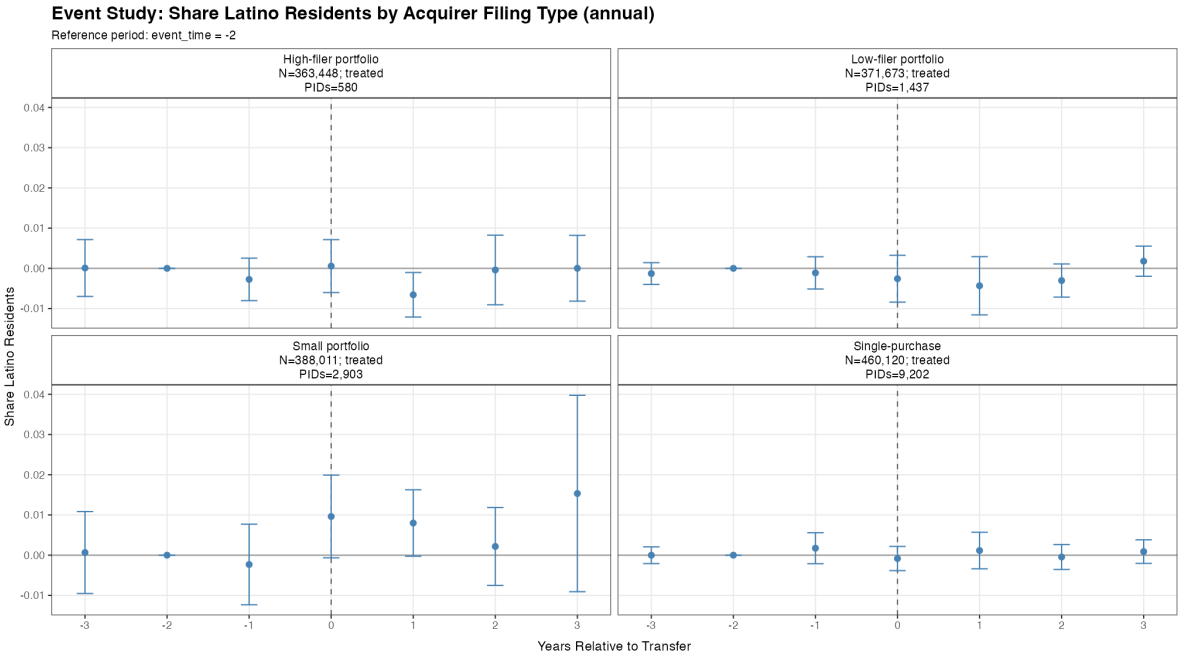

Figure 5: Share Latino residents for 2+ unit buildings, overall and by acquirer filing type.

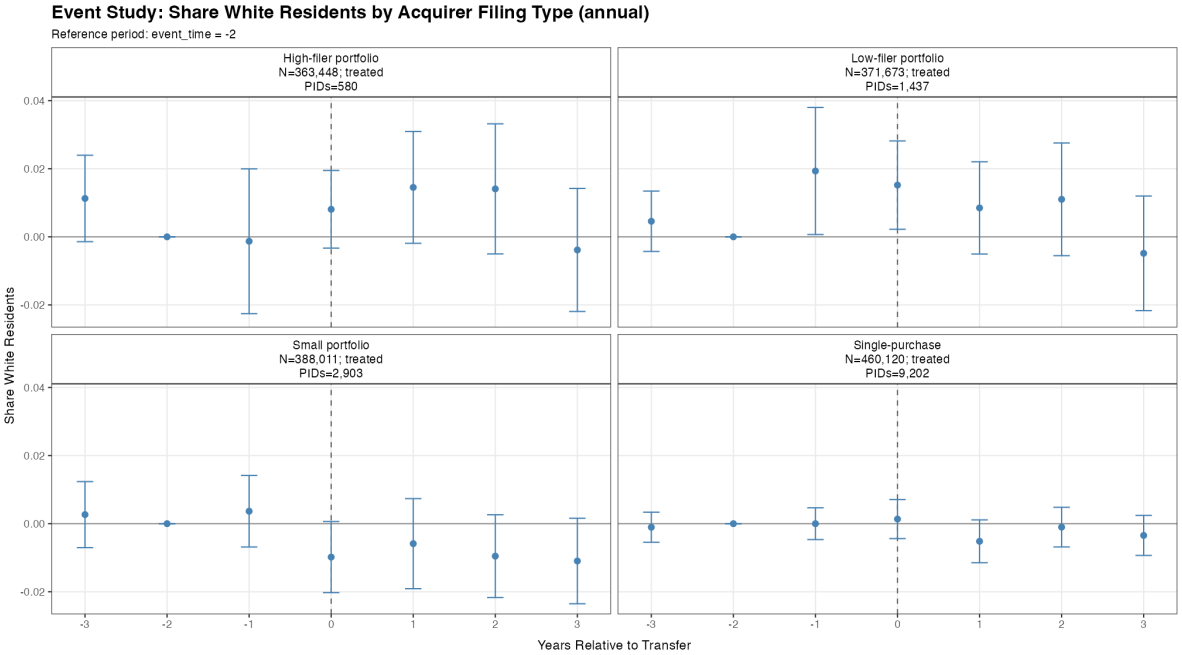

Figure 6: Share White residents for 2+ unit buildings, overall and by acquirer filing type.

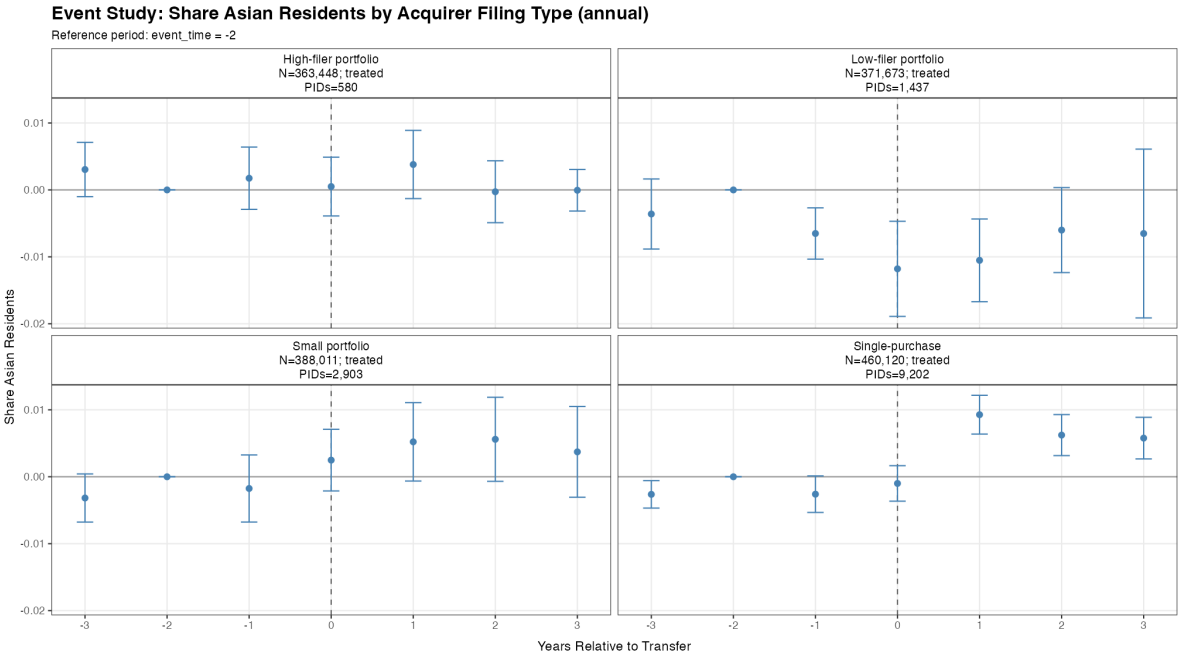

Figure 7: Share Asian residents for 2+ unit buildings, overall and by acquirer filing type.

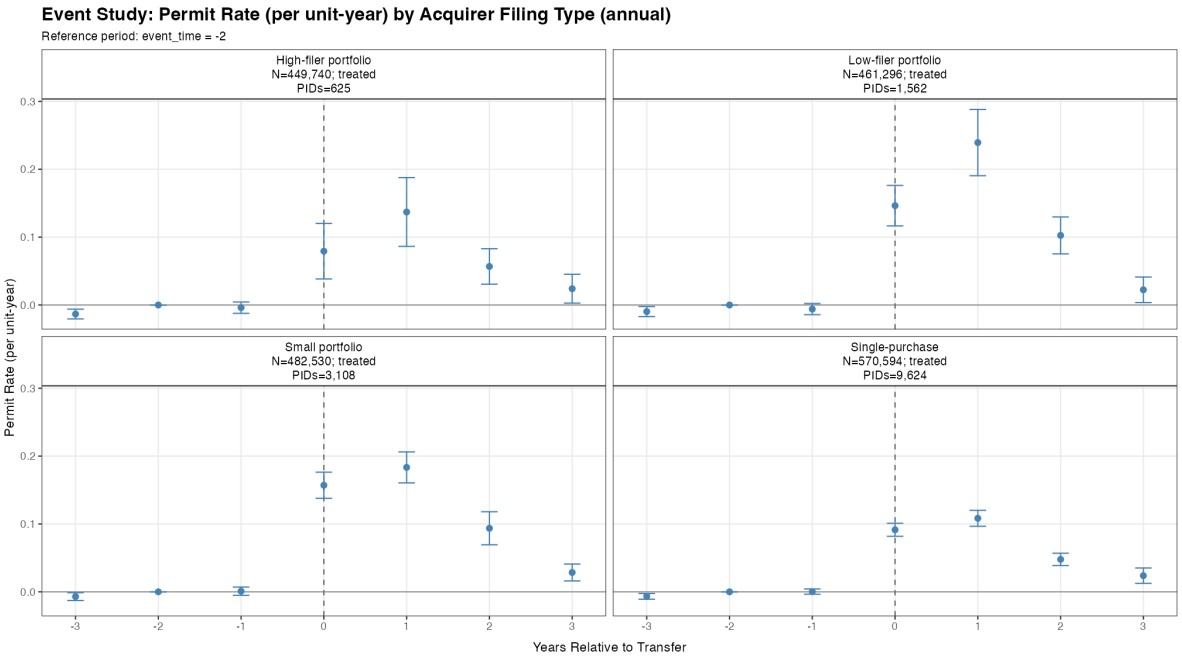

Figure 8: Permit-rate event study for 2+ unit buildings, overall and by acquirer filing type.

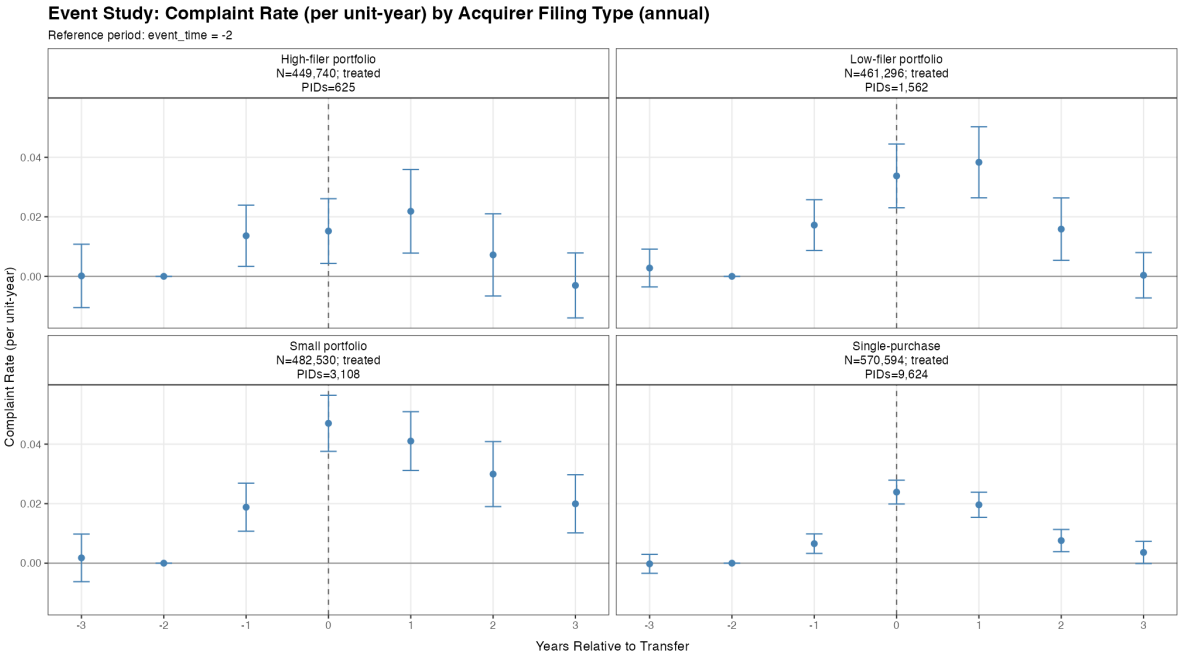

Figure 9: Complaint-rate event study for 2+ unit buildings, overall and by acquirer filing type.

5 Rent Outcomes

The rent analysis is run on all units, but treated buildings are required to have both pre- and post-acquisition support in the relevant rent measure. The report keeps both the unified rent series and the safe single-source series.

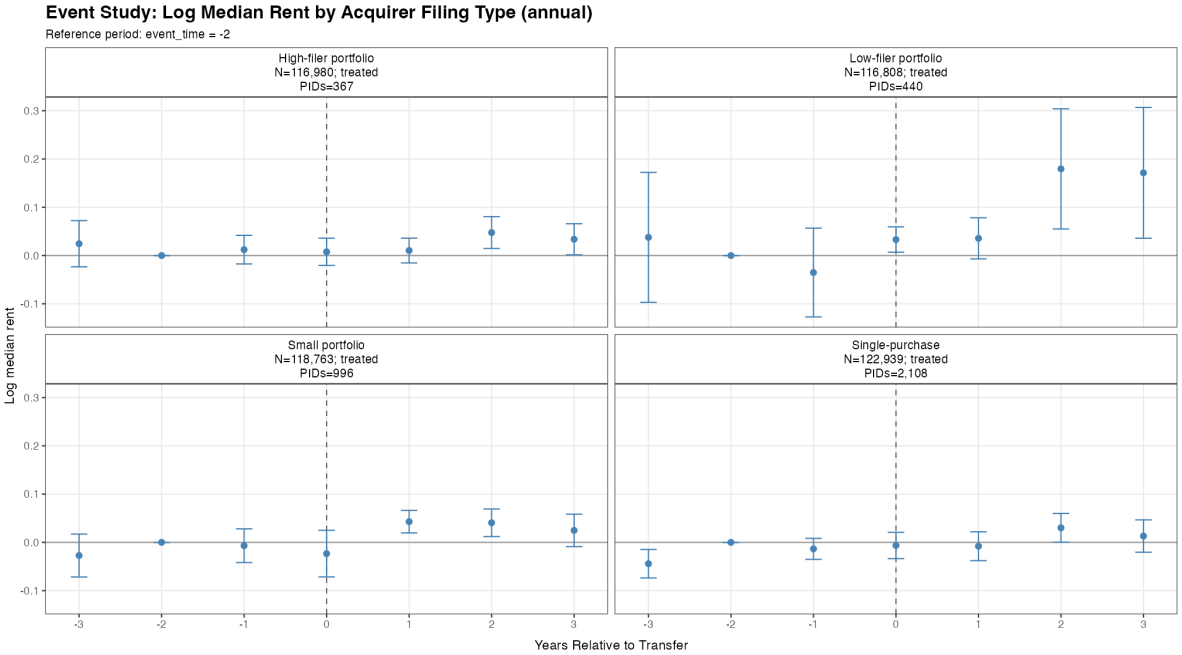

Figure 10: Log median rent event study, all units, overall and by acquirer filing type.

	log_med_rent			
Acquirer type	High-filer portfolio	Low-filer portfolio	Small portfolio	Single-purchase
	(1)	(2)	(3)	(4)
et_m3	0.0245	0.0376	-0.0273	-0.0442***
	(0.0244)	(0.0687)	(0.0227)	(0.0151)
et_m1	0.0122	-0.0352	-0.0069	-0.0135
	(0.0151)	(0.0469)	(0.0178)	(0.0111)
et_0	0.0077	0.0331**	-0.0233	-0.0064
	(0.0144)	(0.0134)	(0.0246)	(0.0139)
et_1	0.0103	0.0357	0.0429***	-0.0080
	(0.0131)	(0.0217)	(0.0119)	(0.0152)
et_2	0.0476***	0.1795***	0.0406***	0.0301**
	(0.0168)	(0.0635)	(0.0146)	(0.0152)
et_3	0.0337**	0.1713**	0.0248	0.0131
	(0.0164)	(0.0691)	(0.0172)	(0.0171)
Observations	116,980	116,808	118,763	122,939
R^2	0.88830	0.88645	0.88769	0.88983
Within R^2	0.00038	0.00411	0.00034	0.00092
PID fixed effects	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓

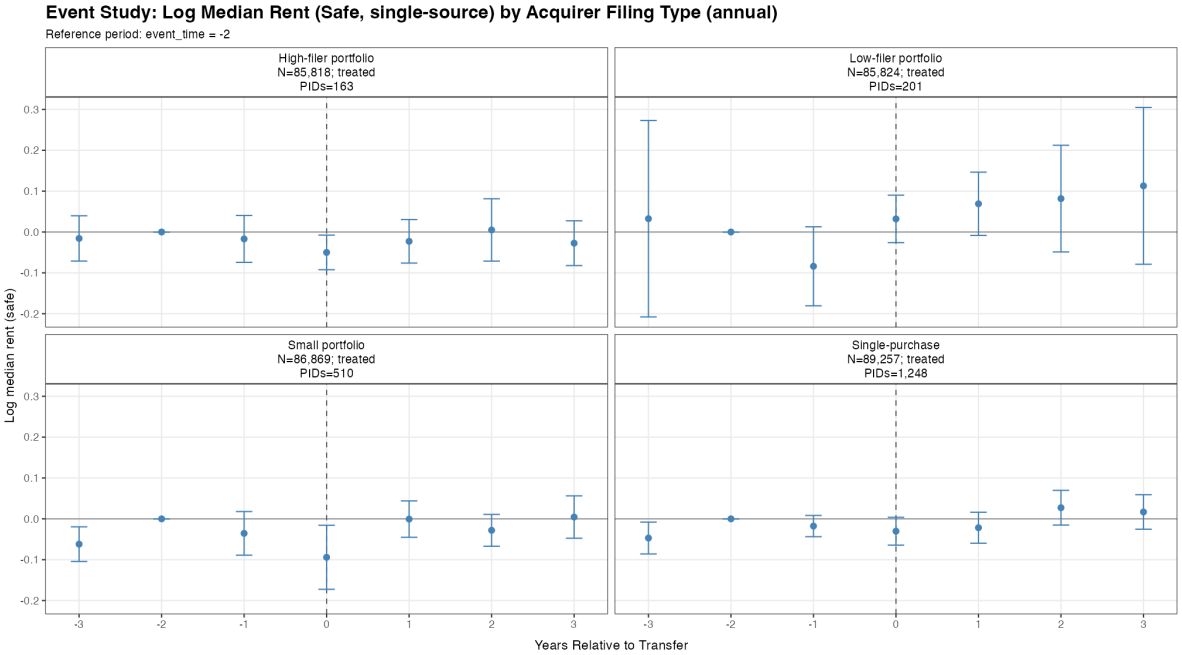

Figure 11: Log median rent (safe single-source series) event study, all units, overall and by acquirer filing type.

	log_med_rent_safe			
Acquirer type	High-filer portfolio	Low-filer portfolio	Small portfolio	Single-purchase
	(1)	(2)	(3)	(4)
et_m3	-0.0157	0.0326	-0.0619***	-0.0469**
	(0.0283)	(0.1226)	(0.0216)	(0.0199)
et_m1	-0.0170	-0.0839*	-0.0355	-0.0177
	(0.0293)	(0.0493)	(0.0272)	(0.0133)
et_0	-0.0500**	0.0321	-0.0940**	-0.0302*
	(0.0216)	(0.0296)	(0.0400)	(0.0174)
et_1	-0.0228	0.0691*	-0.0006	-0.0217
	(0.0272)	(0.0395)	(0.0227)	(0.0193)
et_2	0.0051	0.0817	-0.0281	0.0273
	(0.0389)	(0.0666)	(0.0199)	(0.0217)
et_3	-0.0274	0.1130	0.0044	0.0169
	(0.0279)	(0.0979)	(0.0265)	(0.0215)
Observations	85,818	85,824	86,869	89,257
R^2	0.91663	0.91568	0.91662	0.91800
Within R^2	0.00036	0.00185	0.00074	0.00160
PID fixed effects	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓

6 Rental Evidence

These regressions show the probability of observed rental evidence separately for `1-unit` and `2+ unit` buildings. This is the direct check on the concern that transfers mechanically enter properties into the observed rental market.

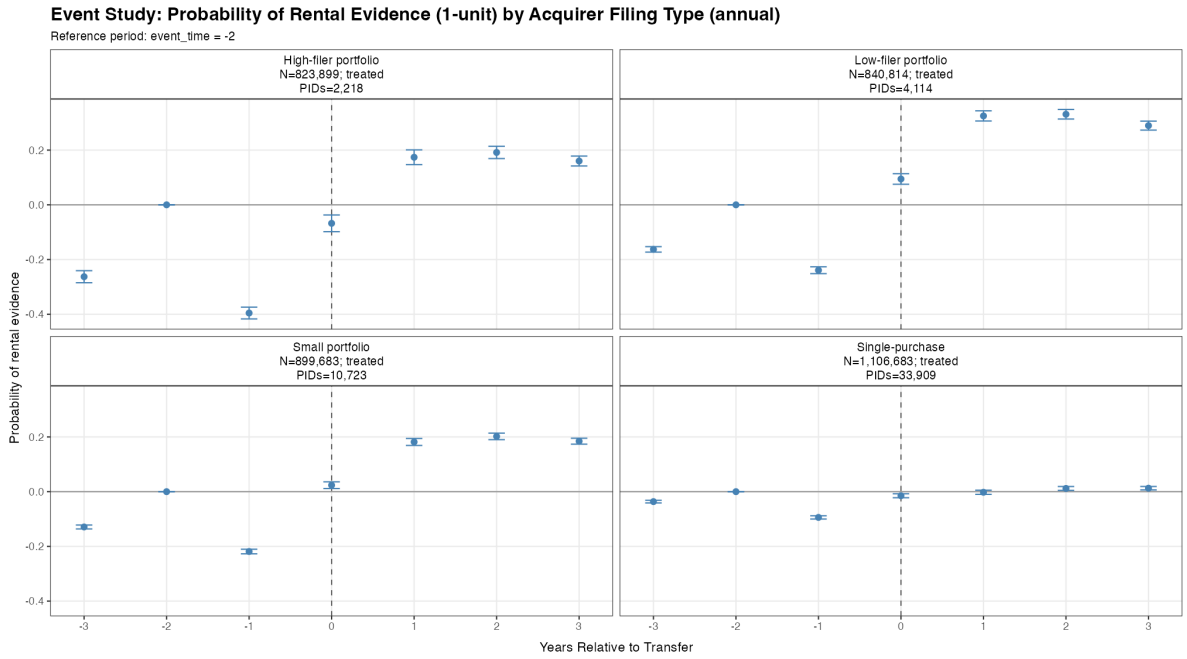

Figure 12: Probability of rental evidence for 1-unit buildings, overall and by acquirer filing type.

	has_rental_evidence			
Acquirer type	High-filer portfolio	Low-filer portfolio	Small portfolio	Single-purchase
	(1)	(2)	(3)	(4)
et_m3	-0.2627***	-0.1628***	-0.1291***	-0.0364***
	(0.0112)	(0.0051)	(0.0037)	(0.0025)
et_m1	-0.3955***	-0.2390***	-0.2187***	-0.0941***
	(0.0110)	(0.0064)	(0.0043)	(0.0030)
et_0	-0.0676***	0.0945***	0.0237***	-0.0149***
	(0.0156)	(0.0098)	(0.0063)	(0.0037)
et_1	0.1739***	0.3250***	0.1817***	-0.0022
	(0.0138)	(0.0095)	(0.0065)	(0.0039)
et_2	0.1915***	0.3311***	0.2020***	0.0118***
	(0.0115)	(0.0089)	(0.0061)	(0.0036)
et_3	0.1601***	0.2895***	0.1845***	0.0127***
	(0.0092)	(0.0084)	(0.0055)	(0.0032)
Observations	823,899	840,814	899,683	1,106,683
R^2	0.65095	0.64885	0.64408	0.62180
Within R^2	0.00477	0.01248	0.01329	0.00169
PID fixed effects	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓

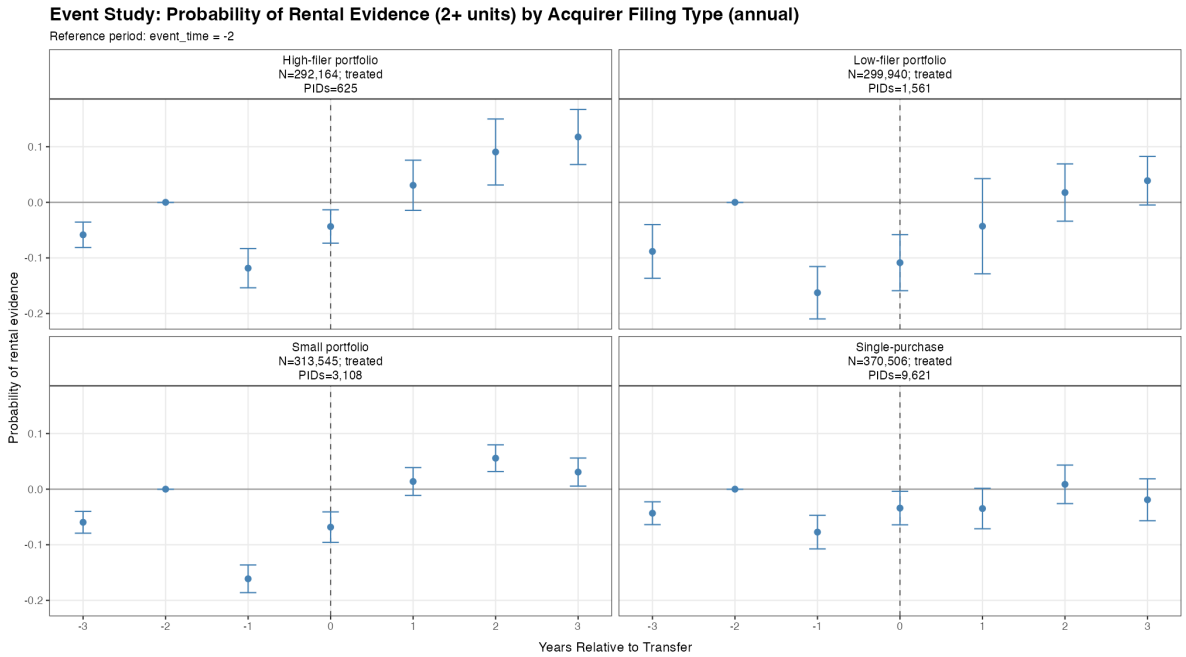

Figure 13: Probability of rental evidence for 2+ unit buildings, overall and by acquirer filing type.

	has_rental_evidence			
Acquirer type	High-filer portfolio	Low-filer portfolio	Small portfolio	Single-purchase
	(1)	(2)	(3)	(4)
et_m3	-0.0584***	-0.0883***	-0.0596***	-0.0433***
	(0.0117)	(0.0246)	(0.0100)	(0.0105)
et_m1	-0.1184***	-0.1626***	-0.1612***	-0.0772***
	(0.0180)	(0.0240)	(0.0127)	(0.0154)
et_0	-0.0435***	-0.1085***	-0.0683***	-0.0340**
	(0.0153)	(0.0257)	(0.0140)	(0.0154)
et_1	0.0307	-0.0430	0.0138	-0.0349*
	(0.0230)	(0.0437)	(0.0128)	(0.0186)
et_2	0.0905***	0.0176	0.0557***	0.0087
	(0.0303)	(0.0263)	(0.0123)	(0.0177)
et_3	0.1175***	0.0389*	0.0307**	-0.0191
	(0.0253)	(0.0223)	(0.0129)	(0.0192)
Observations	292,164	299,940	313,545	370,506
R^2	0.71742	0.71334	0.71156	0.70131
Within R^2	0.00081	0.00157	0.00175	0.00114
PID fixed effects	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓

7 Appendix: Output Inventory

The streamlined annual writeup now depends on the following output families:

- `annual_filing_core_*`
- `annual_filing_acq_2plus_*`
- `annual_occupancy_2plus_*`
- `annual_pct_*_2plus_*`
- `annual_permit_rate_2plus_*`
- `annual_complaint_rate_2plus_*`
- `annual_rent_core_*`
- `annual_rent_safe_core_*`
- `annual_rental_evidence_1unit_*`

- `annual_rental_evidence_2plus_*`

Legacy annual outputs and scratch stacked CSVs are no longer required for this document.